Somaiya Vidyavihar University

K. J. Somaiya School of Engineering

Department of Electronics and Computer Engineering

Multimodal Deepfake Detection for Images, Video and Audio

Submitted In Partial Fulfillment of Requirements
For the Degree Of

Bachelor of Technology

(Electronics and Computer Engineering)

By

Shivansh Gupta Roll No: 16014023052
Vaibhav Shiroorkar Roll No: 16014024805
Adithya Jithesh Roll No: 16014024803

Guide

Dr. Tejashree Phatak
Assistant Professor

Somaiya Vidyavihar University
Vidyavihar, Mumbai - 400 077

2026-2027

Somaiya Vidyavihar University

K. J. Somaiya School of Engineering

Certificate

This is to certify that the project report entitled "**Multimodal Deepfake Detection for Images, Video and Audio**" submitted by the following students at the end of Semester VII of LY B. Tech is a bona fide record for partial fulfillment of requirements for the degree of **Bachelor of Technology (Electronics and Computer Engineering)** of Somaiya Vidyavihar University.

Shivansh Gupta	(16014023052)
Vaibhav Shiroorkar	(16014024805)
Adithya Jithesh	(16014024803)

Guide	**Head of the Department**
Dr. Tejashree Phatak	Dr. Jagannath Nirmal
Examiner	**Director**

Date:

Place: Mumbai-77

Somaiya Vidyavihar University

K. J. Somaiya School of Engineering

Declaration

We declare that this written report submission represents the work done based on our and/or others' ideas with adequately cited and referenced the original source. We also declare that we have adhered to all principles of intellectual property, academic honesty and integrity as we have not misinterpreted or fabricated or falsified any idea/data/fact/source/original work/matter in our submission.

We understand that any violation of the above will be cause for disciplinary action by the college and may evoke the penal action from the sources which have not been properly cited or from whom proper permission is not sought.

Signature of the Student	**Signature of the Student**	**Signature of the Student**
Shivansh Gupta **Roll No.** 16014023052	**Vaibhav Shiroorkar** **Roll No.** 16014024805	**Adithya Jithesh** **Roll No.** 16014024803

Date:

Place: Mumbai-77

Acknowledgements

We would like to express our sincere gratitude to our project guide, Dr. Tejashree Phatak, Assistant Professor, Department of Electronics and Computer Engineering, K. J. Somaiya School of Engineering, for her guidance, advice and encouragement throughout this project.

We thank Dr. Jagannath Nirmal, Head of the Department of Electronics and Computer Engineering, for the facilities that made the work possible, and the faculty and staff of the Department of Electronics and Computer Engineering and of K. J. Somaiya School of Engineering, Somaiya Vidyavihar University, for their support.

We acknowledge the authors of the datasets used in this work, FaceForensics++, Celeb-DF, Celeb-DF++, DeepFakeDetection, ASVspoof 2019, In-the-Wild, LAV-DF, FakeAVCeleb, DeepAction, DF26, Community Forensics and the ForenSynths generated-image sets, who made them available for research, and the developers of the open-source software on which the project is built, among them PyTorch, timm, Hugging Face Transformers, OpenCV and Flask.

Finally we thank our families and friends for their patience and support.

Contents

Certificate i

Declaration ii

Acknowledgements iii

List of Figures v

List of Tables vi

Abbreviations vii

I Introduction 1

II Related Work 1

III System and Method 2

III-A Tasks and score semantics . . . 2

III-B Preprocessing, head, calibration and threshold . . . 2

III-C Encoder sweep and the cost rule . . . 2

IV Evaluation Protocol 2

IV-A Data and group-disjoint splits . . . 2

IV-B Sealed tests, metrics and statistics . . . 3

IV-C Confirmatory tests . . . 3

V Results 4

V-A From the working version to the frozen system (exploratory) . . . 4

V-B Sealed results and confirmatory tests . . . 4

V-C Thresholds and false alarms (post hoc) . . . 5

V-D After the freeze . . . 5

V-E Comparison with published results . . . 6

VI Discussion 7

VII Limitations 7

VIII Conclusion 7

References 8

List of Figures

1 Three scoring paths 3
2 Group-disjoint splits 3
3 Development AUC along the ladder 5
4 LAV-DF per-cell AUC and control 6

List of Tables

I	Encoder sweep outcome	2
II	Datasets and roles	4
III	Development ladder	5
IV	Sealed-test AUC	6
V	Confirmatory tests	6
VI	Steps after the freeze	7

Abbreviations

Term	Expansion
AUC	Area Under the ROC Curve
C2PA	Coalition for Content Provenance and Authenticity
CLIP	Contrastive Language-Image Pre-training
CNN	Convolutional Neural Network
DFD	DeepFakeDetection (dataset)
DFDC	DeepFake Detection Challenge
ECE	Expected Calibration Error
EER	Equal Error Rate
FNR, FPR	False-Negative Rate, False-Positive Rate
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
JPEG	Joint Photographic Experts Group
LA	Logical Access (ASVspoof 2019)
NLL	Negative Log-Likelihood
ROC	Receiver Operating Characteristic
TPR	True-Positive Rate
rPPG	remote Photoplethysmography
SNR	Signal-to-Noise Ratio
SSL	Self-Supervised Learning
T-A, T-I, T-V	speech task, generated-image task, face-video task
ViT	Vision Transformer

Dataset and model names (for example FaceForensics++, Celeb-DF, LAV-DF, DINOv2, WavLM) are proper names and are not expanded.

Multimodal Deepfake Detection for Images, Video and Audio

Shivansh Gupta Vaibhav Shiroorkar Adithya Jithesh
Department of Electronics and Computer Engineering
K. J. Somaiya School of Engineering, Somaiya Vidyavihar University
Vidyavihar, Mumbai - 400 077
Dr. Tejashree Phatak (Guide)
Assistant Professor

***Abstract*: Published deepfake detectors are compared under different protocols. We build a detector for images, videos and the speech in videos as a documented path: one frozen pretrained encoder with a linear head per task, then iterations whose hypotheses and pass lines were written before each run. Splits are group-disjoint, tuning uses a development slice, sealed test sets are read once per frozen iteration, and three paired group-bootstrap tests were declared before that read. On the sealed Celeb-DF family (7 650 videos, 344 groups: 44 identities and 300 single YouTube videos) the final video system, a fine-tuned SigLIP encoder, reaches an AUC of 0.8218 against 0.6238 for the working version (+0.1980 [+0.1688, +0.2242]). It is non-inferior (margin 0.0100) to an Xception baseline re-run at 224 px; the declared 95 % superiority rule was met (+0.0176 [+0.0025, +0.0327]) but does not survive Bonferroni correction over the three tests. On sealed In-the-Wild speech (42 speakers) the final system reaches 0.9148 against 0.7516 (+0.1633 [+0.1177, +0.2266]); part of this gain removes a silence and level shortcut, so it is not all better detection of synthesis. The ProGAN-trained image probe misses 0.7395 of the images of 21 recent generators. On LAV-DF a fused audio-visual score looked near perfect, but its visual part is at chance within the produced videos (0.5032), so no audio-visual detection is claimed. On FakeAVCeleb the pre-stated criteria were met, but codec metadata alone separates real from fake video (0.9985), so they are not evidence of detection quality; thresholds do not transfer.**

***Index Terms*: deepfake detection, multimodal, evaluation protocol, frozen encoders, group bootstrap, sealed test**

I. Introduction

Face swaps, talking-face synthesis, generated images and cloned voices can be produced with consumer hardware and public software [1]. A useful checker has to accept an image, a video or the audio of a video, and say how far its output can be trusted. Published numbers come from different protocols: in one benchmark in which every detector was trained on FaceForensics++ (c23), a plain Xception network reached an average AUC of 0.9450 on the within-domain test sets and 0.7718 on the cross-domain ones [2]. A single in-dataset number therefore says little about new material, and the headline of this work is always a dataset never used in training.

Our goal was a final-year engineering project with two parts of equal weight: a working local system and a record of how it was built. We built the simplest end-to-end detector first, read its diagnostics, and changed one thing at a time, each change starting from a hypothesis, a success line and a failure line written before the run. Failed changes stay in the record.

Contributions: (1) A written evaluation protocol: group-disjoint splits, a development slice separate from sealed test sets, a guarded and logged single reading of each frozen iteration, paired group-bootstrap statistics, and three confirmatory tests declared before any sealed read (Section IV). (2) An encoder selection rule that prefers the cheapest candidate when the evidence does not separate it from the leader, and a measured path of seven experimental iterations with the reason each change was kept or dropped (Sections III and V). (3) Sealed confirmatory results for face video and speech, with the negative results that limit them.

What is not claimed: No audio-visual detection or fused verdict, no pulse-based detection, no generated-video detection, no detection of images from modern generators, no temporal localisation and no Gemini capability. Published numbers are given as context; the only directly comparable row is the re-run Xception baseline.

II. Related Work

Face forgery and generated images: FaceForensics++ reports that detectors lose accuracy as compression increases [3]; Celeb-DF was the hardest of the datasets its authors compared [4], and Celeb-DF++ adds 21 newer methods [5]. DeepfakeBench finds plain backbones close to elaborate methods under shared settings [2], and detectors trained on FaceForensics++ can learn its identities [6], which is why splits must keep identities apart. A linear probe on frozen features of a vision-language model generalised across image generators in its authors' tests [7], and a linear support vector machine on CLIP features trained on few examples of one generator was evaluated on many generators [8]. The protocol of training on ProGAN images comes from a study of CNN-generated images [9]. DF26 benchmarks detectors on recent open and commercial video generators [10].

Speech, audio-visual and physiological cues: ASVspoof 2019 is the standard database of synthesised and converted speech with unseen attacks in its evaluation set [11]; AASIST applies graph attention to the raw waveform [12]; models trained on it degrade strongly on In-the-Wild recordings [13], and frozen self-supervised encoders

with a simple back end transfer well [14]. FakeAVCeleb provides visual and cloned-voice manipulations [15]; LAV-DF regenerates only the edited segments [16]. Pulse signals were proposed as evidence [17], but high-quality fakes can carry a pulse [18] and an independent reproduction reported a much lower AUC [19].

Encoders and calibration: Our candidates are generic pretrained encoders: supervised CNNs [20], [21], self-supervised vision transformers [22], image-text encoders [23], [24], self-supervised speech encoders [25-27] and a weakly supervised recogniser [28]. Transfer accuracy depends on the source model in ways that need measuring [29]. Temperature scaling, one parameter fitted on validation data, is an effective calibration correction on the tasks its authors studied [30].

Gap: Published numbers rarely share data, split and metric, and few studies keep a development slice apart from a test set that is read once. We ask a narrower question: on a protocol fixed in advance, how far does a simple multimodal system get, and what does each change buy?

III. System and Method

A. Tasks and score semantics

Let x be an image, a video or an audio clip and $y \in \{0, 1\}$ its label ($y = 1$: manipulated). There are three tasks: face video (T-V), generated images (T-I) and speech (T-A). A video with an audio track gets a visual and an audio result side by side (Fig. 1); there is no combining rule and no temporal localisation. Detector scores are temperature-scaled on each detector's validation split, which makes them a probability only for content like that split; on other sources their reliable property is their order, and no output is the probability that a file is fake. The frozen detectors are served by a local, eight-module analysis platform that reports one finding per module with its tested scope and no overall score; its design, wording rules and software tests (713 passed, 2 skipped (opt-in GPU tests), 13 deselected (data-dependent tests)) are in the technical report.

B. Preprocessing, head, calibration and threshold

One shared package prepares inputs for training, evaluation and inference. A video gives 16 uniformly spaced frames; the YuNet detector [31] finds the largest face, which is enlarged by 1.3, cropped to 224×224 and passed through a JPEG round trip at quality 95; frames without a face are left out. An image is face-cropped if a face is found and otherwise processed whole by the encoder's own processor. Audio uses the first 64 600 samples at 16 kHz, repeat-padded if shorter. A hash of this configuration is stored in every checkpoint and checked at load. For each task one pretrained encoder ϕ is frozen and a linear head $z = \mathbf{w}^\top \tilde{\phi}(x) + b$ is trained on standardised features with a class-balanced binary cross-entropy, AdamW (learning rate 10^{-3}, weight decay 10^{-4}), early stopping with patience 10 and three head seeds. For a video with face frames F,

$$s(x) = \sigma\left(\frac{\ell(\bar{p}(x))}{T}\right), \quad \bar{p}(x) = \frac{1}{|F|}\sum_{f \in F} \sigma(z_f), \qquad (1)$$

TABLE I
Encoder sweep on the development slices (exploratory): selected encoder and its AUC [95 % group-bootstrap interval].

Task	Tie set	Selected, development AUC
T-V, 15 identities	DINOv2 small and base, SigLIP base, both CLIP	DINOv2 small 0.6933 [0.6639, 0.7314] (leader CLIP OpenAI 0.7464 [0.6721, 0.8306])
T-I, 2 units	CLIP OpenAI only	CLIP ViT-L/14 OpenAI 0.9935 [0.9803, 0.9982]
T-A, 12 speakers	HuBERT, WavLM, Whisper	HuBERT base 0.6100 [0.5434, 0.7256]; inconclusive

with $\ell(q) = \ln \frac{q}{1-q}$ and the temperature T minimising the negative log-likelihood on the validation split. The threshold maximises balanced accuracy on the same calibrated validation scores and is then fixed for every other dataset.

C. Encoder sweep and the cost rule

Seven image encoders (EfficientNet-B4, Xception, DINOv2 small and base, two CLIP ViT-L/14, SigLIP base) and four speech encoders (wav2vec 2.0 base, WavLM base plus, Whisper small, HuBERT base) were scored with features extracted once. Candidates are ranked by mean development AUC; the *tie set* holds every candidate whose paired 95 % group-bootstrap interval against the leader includes zero, and the cheapest member of the tie set wins (extraction time per 1 000 inputs, then parameters). A from-scratch network is reported as a reference and excluded, and DINOv3 was left out because its access needs a manual gate. Table I gives the outcome.

On face video the best non-CNN candidate minus the best supervised CNN is +0.0994 [+0.0228, +0.1723] on this slice, so frozen self-supervised and vision-language features were ahead of supervised CNNs there; the speech sweep verdict is inconclusive (best lower bound 0.5338), and all three slices are small.

The working version (I-0) combines the three selected encoders and a local web page; stored-feature scores and scores computed by the inference code from raw files agree within 0.0019 (tolerance 0.02). Its diagnostics set the next steps. The face-video head reached a training AUC of only 0.8367 (the frozen feature underfits its own training data). The speech head lost 0.384 AUC from validation to development: real In-the-Wild speech was scored as fake, and recording cues such as leading silence and level reverse their sign between the two corpora. Temperature scaling fitted on validation made calibration worse on the development slice for both tasks.

IV. Evaluation Protocol

A. Data and group-disjoint splits

Table II lists the data per role. All datasets are research-only; Celeb-DF, Celeb-DF++ and FakeAVCeleb forbid publishing derived data and DF26 and Community Forensics forbid showing their media, so results here are aggregates. Data is split once by source identity (Fig. 2): the group is the source video with its person, the speaker for speech, the generator

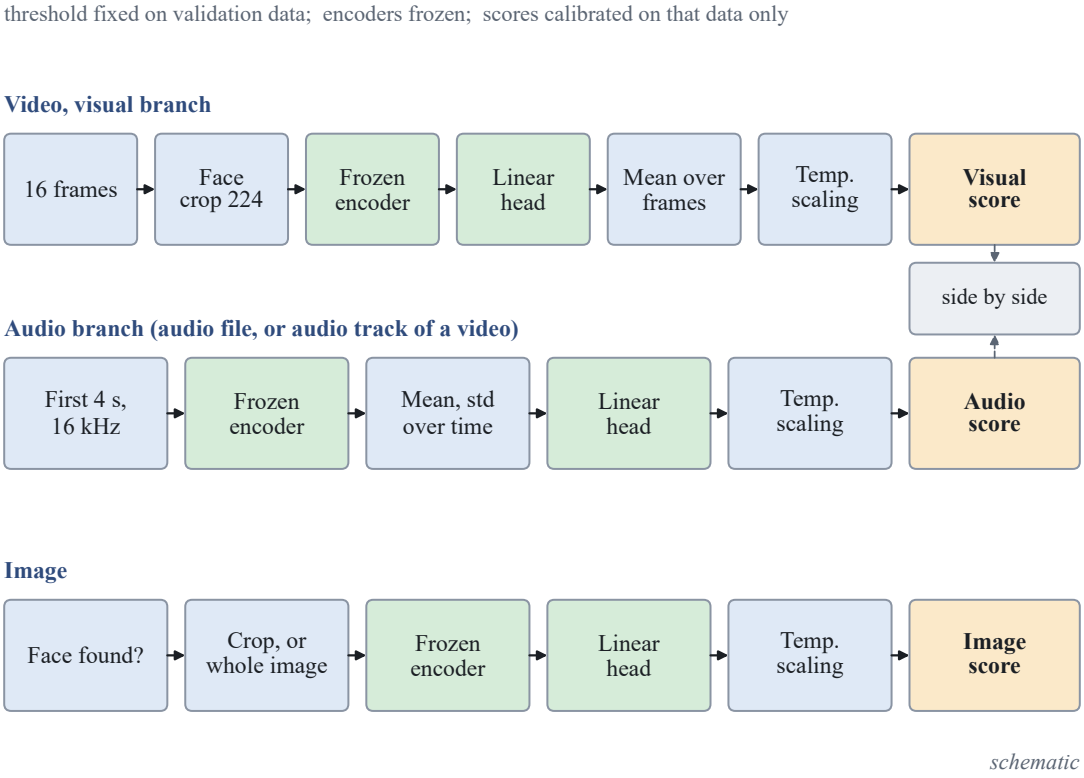


Fig. 1. The three scoring paths of the first version. Encoders are frozen, heads are linear, each score is temperature-scaled on the validation split of its task. Visual and audio scores are not fused.

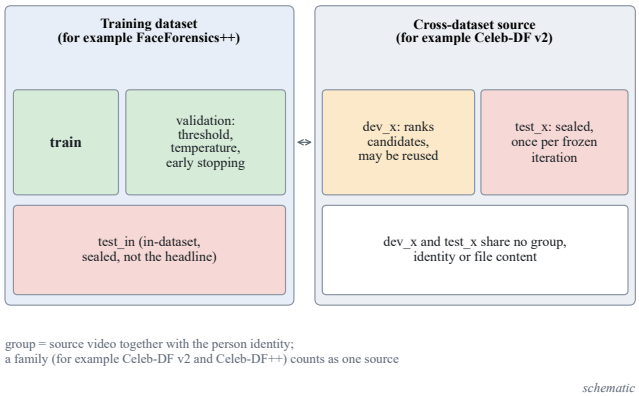


Fig. 2. Schematic of the splits and of the sealed reading. The development slice and the sealed tests share no group, identity or file content with each other or with the training data.

unit for the image development slice, and no group sits on two sides of a split. Celeb-DF++ samples of the 15 development identities and byte-identical duplicates were removed, leaving 744 real and 6 906 fake videos in 344 groups (44 identities and 300 single YouTube videos). A script checks disjointness including a content-hash de-duplication.

B. Sealed tests, metrics and statistics

The development slice ranks candidates and judges iterations; it may be read repeatedly and every verdict on it is exploratory. The sealed sets (`test_in`, `test_x`) are read once per frozen iteration through a guarded command that refuses a sealed split unless the run is flagged final, writes a ledger entry with the score-file hash, and refuses a second reading without a logged reason; the final reading produced 74 ledger entries and no repeat. The metric is the ROC-AUC, $\mathrm{AUC} = P(S_1 > S_0) + \frac{1}{2}P(S_1 = S_0)$ for the scores of a random fake and a random real sample; it uses only ranks and does not depend on the threshold or the class balance. We also report the equal error rate (EER), balanced accuracy at the validation threshold and the expected calibration error with 15 equal-width bins, $\mathrm{ECE} = \sum_b \frac{n_b}{n}|\mathrm{acc}_b - \mathrm{conf}_b|$. Intervals are 95 % percentile intervals from 2 000 bootstrap resamples whose unit is the group, never the clip or frame; with several seeds the metric is averaged over seeds within each resample. Comparisons are paired on the same resampled groups; post-freeze proportions use Wilson intervals. Fine-tuning used one seed per learning rate and only base-size encoders, so that a run fits in 10 GiB of the 12 GB card.

C. Confirmatory tests

Only three tests are confirmatory, declared and committed before the first sealed read: T1, final minus I-0 on the sealed Celeb-DF family (video); T2, the same on sealed In-the-Wild

TABLE II
DATASETS AND ROLES. SEALED SETS WERE READ ONCE PER FROZEN ITERATION.

Task	Train / validation	Development slice	Sealed tests
Face video	FaceForensics++ c23 (four methods), 3 600 / 700 clips	Celeb-DF v2, 1 728 clips, 15 identities	`test_x`: Celeb-DF family (7 650 videos, 344 groups), DFD (3 431, 28 actors); `test_in`: FaceForensics++ (700, 70 groups)
Image	ProGAN, 14 / 6 classes	2 CNN-synthesis units (5 998)	`test_x`: CNN-synthesis (48 805, 16 groups), diffusion (10 000, 10 groups); `test_in`: ProGAN (8 000)
Speech	ASVspoof 2019 LA, 25 380 / 24 844	In-the-Wild, 12 speakers (6 991)	`test_x`: In-the-Wild (24 788, 42 speakers); `test_in`: ASVspoof 2019 LA evaluation (71 237, 67 speakers)
Audio-visual	LAV-DF subset, 500 per cell / 2 000	none	LAV-DF `test_in`, 2 000 videos (1 736 groups)
After the freeze	FakeAVCeleb v1.2 identity split (development half 5 041 clips, sealed half 5 013, 250 identities each); DeepAction (2 605 clips) and DF26 (2 691 clips, 10 generators) for generated video; Community Forensics test-only subset (11 550 images, 21 generators).		

(speech); T3, final video system minus a re-run Xception baseline on the Celeb-DF family. T1 and T2 read "improved" if the lower bound of the paired 95 % interval is above zero. For T3, with L that lower bound, non-inferiority is

$$L > -\delta, \qquad \delta = 0.0100, \tag{2}$$

where δ was set before the sealed read as twice the larger seed standard deviation of the baseline with a floor of 0.01; the floor decides. The declared superiority rule, $L > 0$ at the 95 % level, is a stricter claim than non-inferiority and is read together with the Bonferroni line below. Each test also prints a 98.33 % line (Bonferroni over three, same bootstrap draws). It was declared as a robustness check beside the 95 % decision rule; Bonferroni control holds under any dependence between the three tests, so a reader who wants family-wise control should read the 98.33 % column. Everything else, including every other sealed number, is exploratory.

V. RESULTS

A. From the working version to the frozen system (exploratory)

Table III follows each change from the limitation that triggered it to its measured result on the development slice (three seeds), and Fig. 3 plots the same values. A change enters only if its written success line is met.

Video: The gain over the frozen probe of the same encoder (+0.1700 [+0.1228, +0.2114], 15 identities) isolates fine-tuning; the Xception gain over the working version (+0.1783) is consistent with it but changes the encoder as well. The written prediction of a gain of at most +0.03 was too small. Two further levers were run on request although their triggers were not met to the letter, and neither entered.

Speech: Each of the three speech changes met its success line, and together they raised the development AUC from 0.6100 to 0.8130; the point estimates rise and adjacent intervals overlap. The sealed staircase repeats this (Table IV). The first lever, which removes leading silence and level cues, accounts for 0.061 of the sealed 0.163 gain (0.7516 to 0.8125).

Audio-visual and pulse: The face-video head is below chance on LAV-DF, so fusion was not tested with it. With heads trained on LAV-DF the fused score reached 0.9935, but a control run afterwards showed that the visual head separates the data set's produced videos from its originals, including audio-only fakes whose frames are untouched, and is at chance within the produced videos (0.5070); a visual head trained on produced videos only was no better. The audio head is the only one with evidence above chance inside produced videos (0.8574 on validation). The pulse check found no recoverable signal; a positive control recovered an injected 1.2 Hz sinusoid of relative green amplitude 0.008 in 0.873 of the clips, which shows the pipeline's sensitivity, not that a pulse exists in compressed video (Fig. 4).

B. Sealed results and confirmatory tests

The final system (fine-tuned SigLIP for video, the frozen CLIP probe for images, the three-encoder speech average) was frozen and read once on the sealed sets with seeds 0 to 2. Table IV gives the AUCs and Table V the three tests.

T1 and T2 show that the changes made on the development slices carried over to sealed data never used for any choice. T3 supports exactly this wording and nothing beyond it: the final video system is non-inferior to the re-run Xception at both levels (margin 0.0100); the declared 95 % superiority rule was met but does not survive the Bonferroni correction (98.33 % interval [−0.0011, +0.0355]); the baseline ran at 224 pixels and is one network family. On the newer Celeb-DF++ generators and on DFD, Xception is higher by point estimate. Part of the 0.163 speech gain is the removal of a silence and level shortcut diagnosed on development data (trimming and normalisation account for 0.061; a level cue remains, single-feature RMS AUC 0.135 [0.054, 0.382] on the 12-speaker development slice), so the gain is not all better detection of synthesis. In the training corpus the leading-silence cue has AUC 0.375 on ASVspoof development data against 0.719 on the In-the-Wild development slice, consistent with a sign-flipped shortcut; 12 speakers do not establish it. Sealed video values are below the development values and sealed speech values above them; these are different data and not a before-and-after pair. On the sealed LAV-DF subset the pattern repeats: the visual head trained on LAV-DF scores 0.9921 against the fused label but is at chance within the produced videos (0.5032), where the audio head reaches 0.8480 (exploratory).

TABLE III

DEVELOPMENT LADDER (EXPLORATORY; DEVELOPMENT SLICE, MEAN OF THREE SEEDS, 95 % GROUP-BOOTSTRAP INTERVAL). FACE VIDEO: CELEB-DF V2, 15 IDENTITIES; SPEECH: IN-THE-WILD, 12 SPEAKERS.

Step	Limitation and change	Result	Reading
I-1 video	training AUC 0.8367, frozen feature underfits; full fine-tuning of SigLIP base at 10^{-5}	frozen probe of the same encoder 0.7396 [0.7031, 0.7832], fine-tuned 0.9096 [0.8776, 0.9388]; paired +0.1700 [+0.1228, +0.2114]	supported; higher rates collapsed
Baseline	published family under the same protocol; Xception fine-tuned, same recipe	0.8716 [0.8364, 0.9076]; SigLIP minus Xception +0.0380 [+0.0172, +0.0592]	basis of T3 and of δ
I-1b, I-1c video	validation-to-development gap; augmentation with mixup (I-1b), mean of three fine-tuned encoders (I-1c)	I-1b +0.0219 [+0.0090, +0.0382] on development, −0.0194 [−0.0251, −0.0129] on validation; I-1c +0.0096 [−0.0020, +0.0231]	neither enters: I-1b is a trade-off beyond the allowance 0.0079, I-1c shows no gain
I-2 speech	recording cues reverse sign; trim silence, peak-normalise	0.6723 [0.6126, 0.7584]; +0.0623 [+0.0114, +0.1048]	supported; RMS level cue remains (AUC 0.135 [0.054, 0.382])
I-3 speech	errors on short or saturated clips; noise, low-pass, reverberation, MP3 copies	0.7455 [0.7052, 0.8121]; +0.0732 [+0.0122, +0.1225]	supported; 12 speakers
I-4 speech	single encoder; mean logit of HuBERT, WavLM, Whisper	0.8130 [0.7523, 0.8808]; +0.0675 [+0.0300, +0.0929]	supported; three encoder passes per clip
I-5, I-6 audio-visual	need audio-visual evidence; frozen heads, then heads trained on LAV-DF with logistic fusion	frozen: visual 0.3838 [0.3599, 0.4079], audio 0.6476 [0.6244, 0.6701]; trained: fused 0.9935 [0.9905, 0.9962], but visual within produced videos 0.5070 [0.4771, 0.5361] (post hoc control)	shortcut; no audio-visual detection claim
I-7 pulse	rPPG as candidate cue; signal check on 432 real clips against a shuffled control	0.301 [0.260, 0.346] above control, 0.6 required	fails; no pulse ground truth

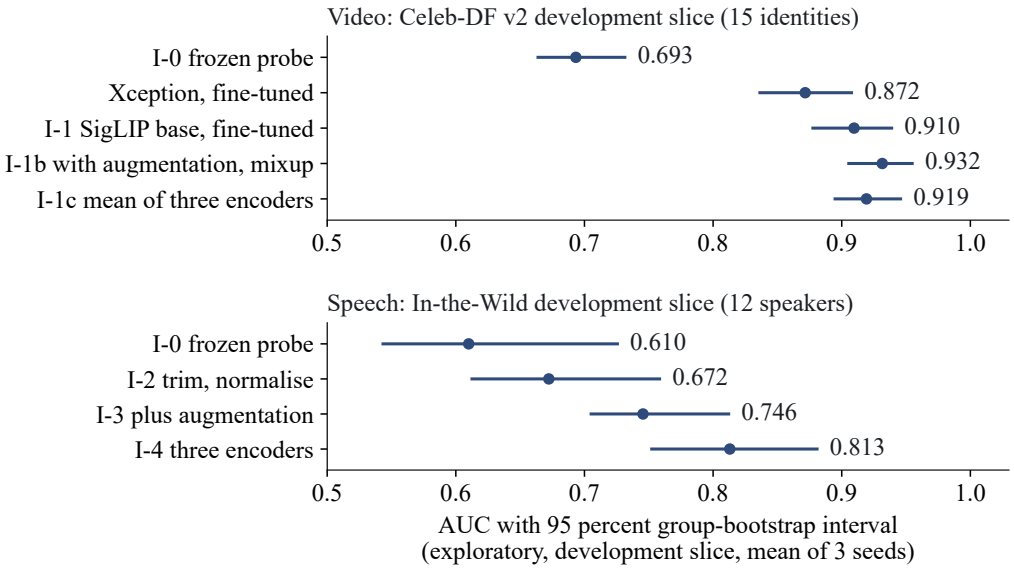


Fig. 3. Development-slice AUC with the 95 % group-bootstrap interval along the ladder (exploratory, mean of three seeds). The widths follow from 15 identities (video) and 12 speakers (speech).

C. Thresholds and false alarms (post hoc)

AUC ignores the threshold, but a user sees a threshold decision. Re-analysing sealed scores already in the ledger, post hoc and not confirmatory, the frozen video threshold flags 0.1098 of the 744 real videos of the Celeb-DF family and 0.1405 of the 363 real DFD videos, so its false alarms differ between test sets; the final speech threshold flags 0.1716 of real In-the-Wild clips. The speech equal error rate is 6.94 % [5.94, 7.79] on ASVspoof 2019 LA and 15.78 % [12.75, 18.70] on In-the-Wild.

D. After the freeze

Further tests were registered with hypotheses and run without changing the frozen detectors; the technical report gives them in full. On the sealed identity half of FakeAVCeleb (5 013 clips) the pre-stated criteria were met (visual 0.9552 [0.9442, 0.9657], audio 0.9263 [0.9149, 0.9366]), but they are

TABLE IV
SEALED-TEST AUC [95 % GROUP-BOOTSTRAP INTERVAL], SEEDS 0 TO 2. ONLY THE CELEB-DF FAMILY AND IN-THE-WILD ROWS FEED A CONFIRMATORY TEST; EVERY OTHER ROW IS SEALED BUT EXPLORATORY.

Sealed set	Items / groups	I-0	Final system	Xception	Role
Celeb-DF family	7 650 / 344	0.6238 [0.6015, 0.6486]	0.8218 [0.8020, 0.8393]	0.8042 [0.7841, 0.8223]	T1, T3
Celeb-DF v2 test part	4 801 / 344	0.6322 [0.6018, 0.6658]	0.8999 [0.8759, 0.9210]	0.8527 [0.8237, 0.8777]	exploratory
Celeb-DF++ fakes, v2 reals	3 593 / 344	0.6119 [0.5821, 0.6427]	0.7105 [0.6877, 0.7331]	0.7351 [0.7083, 0.7599]	exploratory
DFD	3 431 / 28	0.5927 [0.5629, 0.6258]	0.8797 [0.8566, 0.9041]	0.8847 [0.8655, 0.9052]	exploratory
FaceForensics++ c23 (in-dataset)	700 / 70	0.7946 [0.7667, 0.8243]	0.9899 [0.9826, 0.9959]	0.9895 [0.9823, 0.9952]	exploratory
In-the-Wild	24 788 / 42	0.7516 [0.6802, 0.8035]	0.9148 [0.8896, 0.9383]	n/a	T2
after I-2 / I-3			0.8125 / 0.8672	n/a	exploratory
ASVspoof 2019 LA evaluation (in-dataset)	71 237 / 67	0.9684 [0.9564, 0.9777]	0.9835 [0.9798, 0.9872]	n/a	exploratory
Image: CNN-synthesis	48 805 / 16	0.9316 [0.8933, 0.9694]	same head	n/a	exploratory
Image: diffusion	10 000 / 10	0.9354 [0.8967, 0.9665]	same head	n/a	exploratory

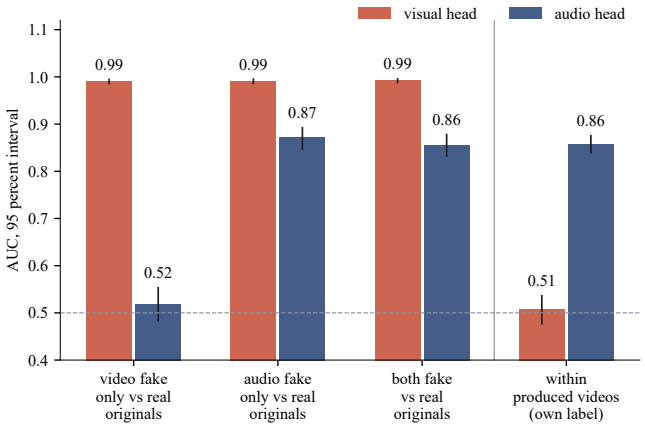


Fig. 4. LAV-DF validation, heads trained on LAV-DF (exploratory): AUC of each head per kind of fake against the real originals, and within the produced videos only (control run afterwards).

TABLE V
THE THREE CONFIRMATORY TESTS: PAIRED AUC DIFFERENCE WITH THE DECLARED 95 % INTERVAL AND THE 98.33 % (BONFERRONI OVER THREE) INTERVAL.

Test	95 % (declared)	98.33 %	Reading
T1 video, final minus I-0	+0.1980 [+0.1688, +0.2242]	[+0.1627, +0.2303]	improved by the declared rule; holds at 98.33 %
T2 speech, final minus I-0	+0.1633 [+0.1177, +0.2266]	[+0.1085, +0.2481]	improved by the declared rule; holds at 98.33 %
T3 video, final minus Xception	+0.0176 [+0.0025, +0.0327]	[−0.0011, +0.0355]	non-inferior at both levels ($\delta = 0.0100$); 95 % superiority rule met, not after Bonferroni; baseline at 224 px

not evidence of detection quality: the video codec coincides with the manipulation method (codec metadata alone separates real from fake video at 0.9985 on the development half), real and fake audio come from different recording channels, and at the frozen thresholds about half of the fake videos are missed (sensitivity 0.480); thresholds do not transfer. The image probe reached a macro AUC of 0.7039 [0.6855, 0.7229] on 21 recent generators and missed 0.7395 [0.7300, 0.7500] of their images at its threshold, part of its separation following the file format. A generated-video head was at chance on DF26 without DF26 training (0.4957 [0.4755, 0.5168]) and flags most real clips from other sources, so it detects a video source rather than generation; a retrained image head ranks better (macro AUC 0.8839, +0.1800 [+0.1643, +0.1959]) but flags 0.5213 of real evaluation images. These experimental heads give no finding (Table VI).

Why the experimental heads were not accepted: A detector enters the product only through an approval table, and the table is empty. The generated-video heads rank their own benchmark but flag most unmodified real clips from other sources (0.7682 and 0.8023 of 440), so they detect a video source, not generation. The retrained image head fails the eligibility line fixed in advance (AUC lower bound at least 0.85, false-alarm upper bound at most 0.10, JPEG drop below 0.10): its false-alarm rate is 0.5213, and re-fitting its threshold on the evaluation images would be test tuning, so it was not done. The whole-image portrait check beats the face path on the same images but misses 0.8063 of generated faces.

Gemini-generated images: No Gemini data was downloaded, no model trained and nothing evaluated; the system has no Gemini capability. The nearest measured proxy is Google Imagen 3, one of the 21 generators above: the frozen head reached AUC 0.7691 [0.7352, 0.8020] and missed 0.8719 of its images, and a PNG-only format score separates the cell at 0.9731, so it is confounded by file format. Imagen 3 is not Gemini. The provider's terms for using outputs to train a detector, provenance, matched real controls and model-version coverage are unresolved, so a registered Gemini study is blocked.

E. Comparison with published results

A published number receives a computed difference only when task, data, protocol and metric match. The only such row is our re-run Xception baseline (T3). Our speech EER on ASVspoof 2019 LA (6.94 % [5.94, 7.79]) is given next to the 1.13 % that the AASIST authors quote [12]; AASIST was not re-run, so this is context, and whether it transfers better was

TABLE VI
STEPS AFTER THE FREEZE; NONE CHANGED A FROZEN DETECTOR. LABELS: SEALED-EXPLORATORY (A HELD-OUT OR SEALED SET READ ONCE, NOT CONFIRMATORY) AND EXPLORATORY (DEVELOPMENT DATA).

Step	Result	Reading
FakeAVCeleb sealed half (sealed-exploratory)	visual 0.9552 [0.9442, 0.9657], audio 0.9263 [0.9149, 0.9366]; visual sensitivity 0.480 at specificity 0.996; audio specificity 0.637	criteria met, not evidence of detection quality; thresholds do not transfer
Codec audit, development half (exploratory)	metadata alone 0.9985; frozen visual 0.9732 before and 0.8772 after a common re-encode (1 000-clip sample, not comparable with the sealed value); codec-matched 0.8856 against 0.8763 (+0.0092 [+0.0006, +0.0180])	codec explains part, not all; fully confounded with method
Generated-video head, DeepAction (DF26 held out once)	DeepAction 0.8281 [0.7908, 0.8662]; DF26 0.4957 [0.4755, 0.5168], closed generators 0.4880 [0.4534, 0.5204]	at chance on DF26; experimental only
Source-confound test (exploratory)	338 of 440 unmodified real clips of FaceForensics++ and LAV-DF flagged, 0.7682 [0.7265, 0.8052], against 0.19 of DeepAction reals	the head separates one library from the rest
Head trained on DF26 open generators (exploratory; closed look sealed-exploratory)	open generators, family left out 0.7453 [0.7174, 0.7710]; closed commercial generators 0.7548 [0.7189, 0.7885] (one counted look; the installed head was at chance there; different real sets); 0.8023 [0.7625, 0.8368] of the 440 real clips flagged	ranking higher, confound persists: not installed
Modern images (sealed-exploratory)	macro AUC 0.7039 [0.6855, 0.7229] on 11 550 images; pooled miss rate 0.7395 [0.7300, 0.7500], false alarms 0.0418 [0.0368, 0.0470]; format-only macro AUC 0.9301	inadequate by the line fixed in advance; format confounded with label
Portrait check (sealed-exploratory)	3 119 face images: whole-image AUC 0.7859 [0.7717, 0.7995], miss rate 0.8063; face path 0.6020; paired +0.1839 [+0.1596, +0.2074]	better than the face path, not usable
Retrained image head (sealed-exploratory)	macro AUC 0.8839 [0.8732, 0.8942]; +0.1800 [+0.1643, +0.1959] over the installed head; false-alarm rate 0.5213 [0.5091, 0.5339]	ranking supported; not eligible
Lip-sync check (exploratory)	false alarms 1.3 % [0.4, 3.8] (old rule 11.7 %) on 230 held-out real clips; Wav2Lip flagged 0 of 160	weak evidence, not a detector

not measured on a common protocol. DeepfakeBench reports a frame-level Xception AUC of 0.7365 on Celeb-DF v2 [2], against our video-level values on a different test subset; no ordering is drawn. On a different test set, with no ranking, the full In-the-Wild set gives 33.9 % in [13] and 7.2 % in [14], against our 15.78 % on 42 of its speakers. On DF26 our head is at chance without DF26 training (0.4957 [0.4755, 0.5168]); the DF26 authors report 44.0 to 69.7 for FaceForensics++-trained detectors without intervals [10]; training data and frame sampling differ, so no ordering is drawn.

VI. DISCUSSION

On the protocol fixed in advance, the changes made on small development slices carried over to sealed data: fine-tuning for face video and three preprocessing and ensembling steps for speech improved on the first version by the declared rule (T1, T2), with the speech caveat above, and against a re-run published network family the video system is non-inferior at both levels. The same evidence shows where the system stops. Scores order clips from sources unlike the training data better than the first version's did, but thresholds fixed on validation data do not transfer, so a flag carries high uncertainty and a score is a ranking, not a probability. That is why the platform of the technical report gives cautious per-module findings and never one number per file.

VII. LIMITATIONS

- *Small slices and groups.* Development slices hold 15 identities, 12 speakers and 2 generator units, so every iteration verdict is exploratory and intervals are wide. Sealed video rests on 344 groups, strongly imbalanced (744 real against 6 906 fake) and mixing two sources; sealed speech on 42 speakers.
- *One sealed run, one baseline.* Each frozen iteration was read once; T3 uses one baseline family at 224 px, δ sits at its floor, the baseline's learning rate lies at the edge of its grid, and T3 does not hold at the Bonferroni level.
- *Speech shortcut.* Part of the speech gain removes a silence and level shortcut; a level cue remains, and 42 speakers are few.
- *Thresholds and false alarms.* Thresholds fixed on validation data do not transfer: the video threshold misses about half of the FakeAVCeleb fakes and flags 0.1098 and 0.1405 of the real videos of the Celeb-DF family and DFD (post hoc), and scores are not probabilities on other sources.
- *Dataset artefacts.* Codec (FakeAVCeleb), file format and size (Community Forensics) and a production cue (LAV-DF) inflate AUC; controls reduce but do not remove them. No audio-visual detection, pulse signal or generated-video detection is claimed.
- *Records and licences.* Some post-freeze registrations entered version control together with their scripts, so their order rests on the project's own record. All data are research-only; the OpenAI CLIP and WavLM weights are used as research-only and the SyncNet weights state no licence.

VIII. CONCLUSION

We built a local multimodal checker and recorded every step that shaped it. A simple first version with frozen encoders showed where it failed; fine-tuning the video encoder and three speech changes addressed the measured weaknesses on devel-

opment data, and on sealed data the frozen system improved on the first version by the declared rule (video +0.1980, speech +0.1633). It is non-inferior to a re-run Xception at both levels, while its 95 % superiority does not survive the Bonferroni correction. Audio-visual fusion, pulse signals, generated-video heads and a retrained image head were tested and not accepted, each for a measured reason, and the frozen thresholds do not transfer. Future work, as recorded in the project plan: scoring the frozen face system on the official Celeb-DF v2 test list beside the baseline, adapting a larger encoder [32], a threshold calibrated on diverse real sources, and the Gemini study once its blockers are resolved.

REFERENCES

[1] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. Canton Ferrer, "The DeepFake detection challenge (DFDC) dataset," arXiv preprint arXiv:2006.07397, 2020.

[2] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, "DeepfakeBench: A comprehensive benchmark of deepfake detection," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023, pp. 4534-4565.

[3] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 1-11.

[4] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for DeepFake forensics," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 3207-3216.

[5] Y. Li, D. Zhu, X. Cui, and S. Lyu, "Celeb-DF++: A large-scale challenging video DeepFake benchmark for generalizable forensics," arXiv preprint arXiv:2507.18015, 2025.

[6] S. Dong, J. Wang, R. Ji, J. Liang, H. Fan, and Z. Ge, "Implicit identity leakage: The stumbling block to improving deepfake detection generalization," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 3994-4004.

[7] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 24 480-24 489.

[8] D. Cozzolino, G. Poggi, R. Corvi, M. Nießner, and L. Verdoliva, "Raising the bar of AI-generated image detection with CLIP," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024, pp. 4356-4366.

[9] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, "CNN-generated images are surprisingly easy to spot... for now," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 8695-8704.

[10] S. Shykula, A. Yermakov, I. Samarskyi, D. Mishkin, J. Cech, and A. Mishchuk, "DF26: We cannot tell fake from real anymore," arXiv preprint arXiv:2609.07369, 2026.

[11] X. Wang, J. Yamagishi, M. Todisco, H. Delgado, A. Nautsch, N. Evans *et al.*, "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, p. 101114, 2020.

[12] J.-w. Jung, H.-S. Heo, H. Tak, H.-j. Shim, J. S. Chung, B.-J. Lee, H.-J. Yu, and N. Evans, "AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 6367-6371.

[13] N. M. Müller, P. Czempin, F. Diekmann, A. Froghyar, and K. Böttinger, "Does audio deepfake detection generalize?" in *Proc. Interspeech*, 2022, pp. 2783-2787.

[14] O. Pascu, A. Stan, D. Oneata, E. Oneata, and H. Cucu, "Towards generalisable and calibrated audio deepfake detection with self-supervised representations," in *Proc. Interspeech*, 2024, pp. 4828-4832.

[15] H. Khalid, S. Tariq, M. Kim, and S. S. Woo, "FakeAVCeleb: A novel audio-video multimodal deepfake dataset," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2021.

[16] Z. Cai, K. Stefanov, A. Dhall, and M. Hayat, "Do you really mean that? content driven audio-visual deepfake dataset and multimodal method for temporal forgery localization," in *Proc. Int. Conf. Digital Image Computing: Techniques and Applications (DICTA)*, 2022, pp. 1-10.

[17] U. A. Ciftci, I. Demir, and L. Yin, "FakeCatcher: Detection of synthetic portrait videos using biological signals," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020, early access.

[18] C. Seibold, E. L. Wisotzky, A. Beckmann, B. Kossack, A. Hilsmann, and P. Eisert, "High-quality deepfakes have a heart!" *Frontiers in Imaging*, vol. 4, 2025.

[19] O. Harraq and T. Aldwairi, "Physiological signals as a forensic modality for talking-face deepfake detection," arXiv preprint arXiv:2607.21776, 2026.

[20] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 1251-1258.

[21] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Machine Learning (ICML), PMLR 97*, 2019, pp. 6105-6114.

[22] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.

[23] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Machine Learning (ICML), PMLR 139*, 2021, pp. 8748-8763.

[24] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 11 975-11 986.

[25] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.

[26] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai *et al.*, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451-3460, 2021.

[27] S. Chen, C. Wang, Z. Chen *et al.*, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505-1518, 2022.

[28] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. 40th Int. Conf. Machine Learning (ICML), PMLR 202*, 2023, pp. 28 492-28 518.

[29] S. Kornblith, J. Shlens, and Q. V. Le, "Do better ImageNet models transfer better?" in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 2661-2671.

[30] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Machine Learning (ICML), PMLR 70*, 2017, pp. 1321-1330.

[31] W. Wu, H. Peng, and S. Yu, "YuNet: A tiny millisecond-level face detector," *Machine Intelligence Research*, vol. 20, no. 5, pp. 656-665, 2023.

[32] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Int. Conf. Learning Representations (ICLR)*, 2022.